

# P8109 Statistical Inference - Comprehensive Review

Lectures 1-10: COMPLETE Theorems, Definitions, and Remarks

Spring 2026

## Essential Exam Strategy

1. **Identify the Support:** Always check if the support  $\mathcal{X}$  depends on  $\theta$  (e.g.,  $0 < x < \theta$ ). If it does, the regularity conditions for CRLB and Fisher Information often fail, and the distribution is NOT an Exponential Family. In such cases, MLE is typically an order statistic ( $X_{(n)}$  or  $X_{(1)}$ ).
2. **Finding UMVUE (The Standard Procedure):**
  - **Step 1:** Find a sufficient statistic  $T$  using the Fisher-Neyman Factorization Theorem.
  - **Step 2:** Show that  $T$  is complete (typically by showing it is the natural sufficient statistic of a full-rank Exponential Family).
  - **Step 3:** Either find a function  $g(T)$  such that  $E_\theta[g(T)] = \tau(\theta)$  (using Lehmann-Scheffé Theorem), OR take any unbiased estimator  $d(\mathbf{X})$  and compute  $E[d(\mathbf{X})|T]$  (Rao-Blackwellization).
3. **Asymptotic Variance:** Before using the Delta Method, check if  $g'(\theta) = 0$ . If it is, the first-order Delta Method yields a degenerate distribution (0 variance), and you must use the second-order Delta Method, which leads to a  $\chi^2_1$  distribution.
4. **Hypothesis Testing Strategy:**
  - **Simple vs. Simple:** Use the **Neyman-Pearson Lemma**. The MP test is always a Likelihood Ratio Test.
  - **One-Sided Composite:** Check for the **Monotone Likelihood Ratio (MLR)** property. If it exists (e.g., in Exponential Families), the UMP test is based on the sufficient statistic  $T(\mathbf{X})$ .
  - **Discrete Distributions:** Remember to include the **randomization constant**  $\gamma$  to achieve an exact size  $\alpha$  test.

## Lecture 1: Normal and Associated Distributions

### 1.1 The Normal Distribution $N(\mu, \sigma^2)$

- **Probability Density Function (1.1):**

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0$$

- **Moment-Generating Function (1.2):**  $M_X(t) = E[e^{tX}] = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$ .
- **Standardization:** If we define  $Z = \frac{X-\mu}{\sigma}$ , then  $Z \sim N(0, 1)$  with density  $f_Z(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ .
- **Theorem 1.1:** If  $X \sim N(\mu, \sigma^2)$ , then for any constants  $a$  and  $b$ ,  $aX + b \sim N(a\mu + b, a^2\sigma^2)$ .
- **Theorem 1.2:** Let  $X_1, \dots, X_n$  be independent random variables with  $X_i \sim N(\mu_i, \sigma_i^2)$ . Let  $W = \sum_{i=1}^n a_i X_i + b$  and  $V = \sum_{i=1}^n c_i X_i + d$ . Then:
  - $W \sim N(\sum a_i \mu_i + b, \sum a_i^2 \sigma_i^2)$ .
  - $W$  and  $V$  are independent **if and only if**  $\sum_{i=1}^n a_i c_i = 0$ .
  - **Remark:** The converse of the independence property is not true for non-normal distributions. This specific independence of linear combinations characterizes the normal distribution.

### 1.2 Associated Distributions $(\chi^2, t, F)$

- **The  $\chi^2$  Distribution:** Let  $Z_1, \dots, Z_k$  be i.i.d.  $N(0, 1)$  random variables. Then  $U = \sum_{i=1}^k Z_i^2 \sim \chi_k^2$ .
  - $\chi_k^2$  is a special case of  $Gamma(\alpha = k/2, \beta = 2)$ .

- $E[U] = k$ ,  $Var(U) = 2k$ .
- MGF:  $M_U(t) = (1 - 2t)^{-k/2}$  for  $t < 1/2$ .

- **The  $t$ -distribution:** Let  $Z \sim N(0, 1)$  and  $V \sim \chi_k^2$  be independent random variables. Then:

$$T = \frac{Z}{\sqrt{V/k}} \sim t_k$$

- $E[T] = 0$  for  $k > 1$ ;  $Var(T) = \frac{k}{k-2}$  for  $k > 2$ .
- As  $k \rightarrow \infty$ ,  $t_k \xrightarrow{d} Z \sim N(0, 1)$ .

- **The  $F$ -distribution:** If  $U \sim \chi_m^2$  and  $V \sim \chi_n^2$  are independent, then:

$$F = \frac{U/m}{V/n} \sim F_{m,n}$$

- **Result (a):** If  $T \sim t_k$ , then  $T^2 \sim F_{1,k}$ .
- **Result (b):** If  $X \sim F_{m,n}$ , then  $1/X \sim F_{n,m}$ .
- **Result (c):** If  $X \sim F_{m,n}$ , then  $E[X] = \frac{n}{n-2}$  for  $n > 2$ .

### 1.3 The Central Limit Theorem (CLT)

- **Theorem 1.3 (Lindeberg-Lévy CLT):** Let  $X_1, X_2, \dots, X_n$  be i.i.d. random variables with  $E[X_i] = \mu$  and  $Var(X_i) = \sigma^2 < \infty$ . Define  $S_n = X_1 + \dots + X_n$ . Then, as  $n \rightarrow \infty$ :

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z \sim N(0, 1)$$

### 1.4 Sampling from Normal Distributions

- **Theorem 1.4:** Let  $X_1, \dots, X_n$  be i.i.d. from  $N(\mu, \sigma^2)$ . Let  $\bar{X}$  and  $S^2$  be the sample mean and sample variance, respectively. Then:

- (a)  $\bar{X}$  and  $S^2$  are independent.
- (b)  $\bar{X} \sim N(\mu, \sigma^2/n)$ .
- (c)  $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ .

- **Corollary 1.1:**

- (a)  $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ .
- (b)  $E[S^2] = \sigma^2$ ,  $Var(S^2) = \frac{2\sigma^4}{n-1}$ .

- **Remarks:**

- (i)  $E(S^2) = \sigma^2$ , which means  $S^2$  is an unbiased estimator of  $\sigma^2$ .
- (ii) If  $\sigma^2$  is estimated by the MLE  $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ , then  $E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2$ , which is biased (it underestimates  $\sigma^2$ ).
- (iii) Since  $Var(S^2) = E(S^2) - [E(S)]^2 > 0$ , it follows that  $E(S) < \sigma$ . Thus,  $S$  is a biased estimator for  $\sigma$  (by Jensen's inequality).

### 1.5 The Weak Law of Large Numbers (WLLN)

- **Theorem:** Let  $X_1, \dots, X_n$  be i.i.d. random variables with finite mean  $E[X_i] = \mu$  and finite variance  $Var(X_i) = \sigma^2 < \infty$ . Then, the sample mean  $\bar{X}_n$  converges in probability to the population mean  $\mu$ :

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu, \quad \text{as } n \rightarrow \infty$$

- **Proof Hint (Markov's):**  $P(|\bar{X}_n - \mu| > \epsilon) = P((\bar{X}_n - \mu)^2 > \epsilon^2) \leq \frac{E(\bar{X}_n - \mu)^2}{\epsilon^2} = \frac{Var(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$ .

# Lecture 2: Exponential Family & Asymptotics

## 2.1 Exponential Family Distributions

- **Definition:** A one-parameter family of densities  $f(x|\theta)$  belongs to the exponential family if it can be expressed as:

$$f(x|\theta) = h(x) \exp\{c(\theta)T(x) - A(\theta)\}$$

where  $h, T$  are known functions of  $x$ , and  $c, A$  are known functions of  $\theta$ .

- **Natural Form (2.1):** Let  $\eta = c(\theta)$  be the natural parameter. The density is:

$$f(x|\eta) = h(x) \exp\{\eta T(x) - A(\eta)\}$$

- **General Exponential Family:** If  $\theta = (\theta_1, \dots, \theta_k)^T$ , the distribution belongs to the EF if:

$$f(\mathbf{x}|\theta) = h(\mathbf{x}) \exp\left\{\sum_{j=1}^k c_j(\theta)T_j(\mathbf{x}) - A(\theta)\right\}$$

- **Full Rank vs Curved:** If the set  $\{c(\theta) : \theta \in \Theta\}$  contains a  $k$ -dimensional open ball in  $\mathbb{R}^k$ , the distribution is said to have **full rank**. Otherwise, it is a curved exponential family.
- **Theorem 2.1:** Let  $\mathbf{X}$  follow the natural form distribution. Then:
  1.  $T(\mathbf{X})$  also belongs to the exponential family.
  2. The MGF of  $T$  is  $M_T(\mathbf{t}) = \exp\{A(\eta + \mathbf{t}) - A(\eta)\}$ .
  3.  $E[T] = \nabla A(\eta)$  and  $Var(T) = \nabla^2 A(\eta)$  (the Hessian).

## 2.2 Common Probabilistic Inequalities

1. **Jensen's Inequality:** Let  $\varphi$  be a convex function. Then  $\varphi(E[X]) \leq E[\varphi(X)]$ . For strictly convex  $\varphi$ , equality holds iff  $X = E[X]$  a.s.
2. **Hölder's Inequality:** Let  $p, q \in [1, \infty]$  such that  $p^{-1} + q^{-1} = 1$ . Then  $E|XY| \leq (E|X|^p)^{1/p} (E|Y|^q)^{1/q}$ .
  - **Cauchy-Schwarz:** When  $p = q = 2$ ,  $E|XY| \leq \sqrt{E[X^2]E[Y^2]}$ .
3. **Markov's Inequality:** Let  $X$  be a random variable. For any  $a > 0$  and  $r > 0$ :

$$P(|X| \geq a) \leq \frac{E|X|^r}{a^r}$$

4. **Chebyshev's Inequality:** Let  $X$  be a random variable with  $E[X] = \mu$  and  $Var(X) = \sigma^2$ . For any  $a > 0$ :

$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

## 2.3 Types of Convergence

- **Definition 2.1:**

1.  $X_n$  **converges in probability** to  $X$  ( $X_n \xrightarrow{P} X$ ) if for any  $\epsilon > 0$ ,  $P(|X_n - X| > \epsilon) \rightarrow 0$  as  $n \rightarrow \infty$ .
2.  $X_n$  **converges in distribution** to  $X$  ( $X_n \xrightarrow{d} X$ ) if  $F_{X_n}(x) \rightarrow F_X(x)$  at all continuity points  $x$  of  $F_X$ .

- **Theorem 2.2:**

1.  $X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X$ .
2.  $X_n \xrightarrow{d} c$  (where  $c$  is a constant)  $\implies X_n \xrightarrow{P} c$ .
3.  $X_n \xrightarrow{d} X$  iff  $E[g(X_n)] \rightarrow E[g(X)]$  for all bounded continuous functions  $g$ .
4. **Lévy's Continuity Theorem:**  $X_n \xrightarrow{d} X$  iff  $M_{X_n}(t) \rightarrow M_X(t)$  for all  $t$  in a neighborhood of 0.

- **Theorem 2.3 (Continuous Mapping Theorem):** Let  $g$  be a continuous function.

1. If  $X_n \xrightarrow{P} X$ , then  $g(X_n) \xrightarrow{P} g(X)$ .

2. If  $X_n \xrightarrow{d} X$ , then  $g(X_n) \xrightarrow{d} g(X)$ .
- **Theorem 2.4:** If  $X_n \xrightarrow{P} X$  and  $Y_n \xrightarrow{P} Y$ , and  $h(x, y)$  is a continuous function, then  $h(X_n, Y_n) \xrightarrow{P} h(X, Y)$ .
- **Theorem 2.5 (Slutsky's Theorem):** Assume  $X_n \xrightarrow{d} X$  and  $Y_n \xrightarrow{P} c$  (constant). Then:
  1.  $X_n + Y_n \xrightarrow{d} X + c$
  2.  $X_n Y_n \xrightarrow{d} cX$
  3.  $X_n/Y_n \xrightarrow{d} X/c$ , provided  $c \neq 0$ .

## 2.4 Delta Method

- **Theorem 2.6 (Delta Method):** Assume  $a_n(\hat{\theta}_n - \theta) \xrightarrow{d} W$ , where  $a_n \rightarrow \infty$  (usually  $a_n = \sqrt{n}$ ). This condition implies  $\hat{\theta}_n \xrightarrow{P} \theta$ . Let  $g(t)$  be a differentiable function with  $g'(\theta)$  existing.
  - **1st Order:** If  $g'(\theta) \neq 0$ , then  $a_n(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} g'(\theta)W$ .
  - **2nd Order (Remark):** If  $g'(\theta) = 0$  but  $g''(\theta) \neq 0$ , then  $a_n^2(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} \frac{g''(\theta)}{2}W^2$ .
  - **Example 2.6:** If  $W \sim N(0, \sigma^2)$ , then  $a_n^2(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} \frac{g''(\theta)}{2}\sigma^2\chi_1^2$ .

## Lecture 3: Parameter Estimations

### 3.1 Method of Moments (MoM)

- **The Equation:** Setting the  $r$ -th sample moment  $m_r = \frac{1}{n} \sum X_i^r$  equal to the  $r$ -th population moment  $\mu_r = E[X^r]$ . The MoM proceeds by setting  $m_r = \mu_r$  for  $r = 1, \dots, k$  to solve for  $k$  parameters.
- **Justification:**  $E(m_r) = \mu_r$ .
- **Remark 3.1:** MoM estimators can be biased. For  $N(\mu, \sigma^2)$ , MoM for  $\sigma^2$  is  $\frac{1}{n} \sum (X_i - \bar{X})^2$ , which is biased.
- **Remark 3.2:** MoM may not exist if population moments don't exist (e.g., Cauchy). However, if we allow  $r$  to be a positive non-integer (e.g.,  $r = 1/2$ ), a MoM estimator can sometimes be obtained.

### 3.2 Maximum Likelihood Estimation (MLE)

- **Likelihood Function (3.2):**  $L(\theta|\mathbf{X}) = f(\mathbf{x}; \theta)$ . For i.i.d. samples,  $L(\theta|\mathbf{X}) = \prod f(x_i; \theta)$ .
- **MLE Definition (3.3):**  $\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta|\mathbf{X})$ .
- **Invariance Principle:** If  $\hat{\theta}$  is the MLE of  $\theta$ , then for any function  $\tau(\theta)$ , the MLE of  $\tau(\theta)$  is  $\tau(\hat{\theta})$ .
- **Remark 3.3 (Important):** Differentiation may not work if the support of  $X$  depends on  $\theta$ . Example: For  $Uniform(0, \theta)$ ,  $L(\theta|\mathbf{X}) = \theta^{-n} I(X_{(n)} \leq \theta)$ . This is monotonically decreasing for  $\theta \in [X_{(n)}, \infty)$ , so  $\hat{\theta} = X_{(n)}$ .
- **Remark 3.4:** MLE may not be unique. For  $Uniform(\theta, \theta + 1)$ , any value in  $[X_{(n)} - 1, X_{(1)}]$  is an MLE.
- **Remark 3.5:** MLE may not exist for some values of  $X$  (e.g., Hypergeometric with  $X = 0$ ).

## Lecture 4: Assessing the Performance of Estimators

### 4.1 Unbiasedness and MSE

- **Definition 4.1 (Bias):** Suppose  $\tau(\theta)$  is a function of  $\theta$ , and let  $T = T(\mathbf{X})$  be an estimator of  $\tau(\theta)$ . Then the bias of  $T$  is defined by  $Bias(T) = E[T] - \tau(\theta)$ .  $T$  is unbiased if  $Bias(T) = 0$ .
- **Definition 4.2 (UMVUE):**  $T^*$  is the Uniformly Minimum Variance Unbiased Estimator if  $E[T^*] = \tau(\theta)$  and  $Var(T^*) \leq Var(T)$  for all  $\theta$ , where  $T$  is any other unbiased estimator.
- **Theorem 4.1:** The UMVUE is unique.
- **Mean Squared Error (4.2):**  $MSE(T) = E[(T - \tau(\theta))^2] = Var(T) + [Bias(T)]^2$ .

## 4.2 Consistency

- **Definition 4.3:**  $T_n$  is a consistent estimator of  $\tau(\theta)$  if  $T_n \xrightarrow{P} \tau(\theta)$  as  $n \rightarrow \infty$ .
- **Theorem 4.2:** If  $\text{Bias}(T_n) \rightarrow 0$  and  $\text{Var}(T_n) \rightarrow 0$  as  $n \rightarrow \infty$ , then  $T_n$  is consistent.
- **Remark to Theorem 4.2:** These conditions are sufficient but not necessary. (An estimator can have infinite variance and still be consistent).

## 4.3 Fisher Information and CRLB

- **Definition 4.4:** Let  $\ell(\theta) = \log L(\theta|\mathbf{X})$ . The Score function is  $S(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$ . The Fisher Information is  $i(\theta) = \text{Var}(S(\theta))$ .
- **Theorem 4.3:** If derivatives and integrals can be switched:

- (a)  $E[S(\theta)] = 0$ .
- (b)  $i(\theta) = E[S(\theta)S(\theta)^T] = -E\left[\frac{\partial^2}{\partial \theta^2} \ell(\theta)\right]$ .

- **Theorem 4.4 (Cramér-Rao Inequality):** Let  $T$  be an unbiased estimator of  $\tau(\theta)$ . Then:

$$\text{Var}(T) \geq \frac{[\tau'(\theta)]^2}{i(\theta)} = \frac{[\tau'(\theta)]^2}{ni_1(\theta)}$$

where  $i(\theta)$  is the information in the whole sample  $\mathbf{X}$  (i.e.,  $I_n$ ), and  $i_1(\theta)$  is the information in a single observation.

- **Definition (MVBU):** Estimators which achieve the CRLB are said to be **efficient** or **minimum variance bound unbiased (MVBU)**. An MVBU is always a UMVUE, but the converse is not necessarily true (some UMVUEs have variances exceeding the CRLB).
- **Equality Condition:** Equality holds iff  $S(\theta) = a(\theta)[T(\mathbf{X}) - \tau(\theta)]$ , which happens iff the distribution is in the Exponential Family.
- **Asymptotic Efficiency (Definition 4.5/Theorem 4.5):** An estimator  $\hat{\theta}$  is asymptotically efficient if:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{i_1(\theta)}\right)$$

This implies that for large  $n$ ,  $\text{Var}(\hat{\theta}) \approx \frac{1}{ni_1(\theta)} = \frac{1}{I_n(\theta)}$ . Under regularity conditions, the MLE is asymptotically efficient.

## Lecture 5: Data Reduction Principles: Sufficiency

### 5.1 Sufficient Statistics

- **Definition 5.1:** A statistic  $T(\mathbf{X})$  is sufficient for  $\theta$  if the conditional distribution of  $\mathbf{X}$  given  $T$  does not depend on  $\theta$ .
- **Theorem 5.1 (Fisher-Neyman Factorization):**  $T(\mathbf{X})$  is sufficient for  $\theta$  if and only if the likelihood can be factored as:

$$L(\theta|\mathbf{x}) = g(T(\mathbf{x})|\theta)h(\mathbf{x})$$

where  $g$  depends on  $\mathbf{x}$  only through  $T(\mathbf{x})$ , and  $h$  does not depend on  $\theta$ .

- **Theorem 5.2:** If  $T$  is sufficient and  $T$  is a function of  $S$  (i.e.,  $T = R(S)$ ), then  $S$  is also sufficient. (Larger statistics preserve sufficiency).

### 5.2 Rao-Blackwell Theorem

- **Theorem 5.3:** Let  $T$  be a sufficient statistic and  $d(\mathbf{X})$  be an unbiased estimator of  $\tau(\theta)$ . Define  $d^*(T) = E[d(\mathbf{X})|T]$ . Then:
  - (a)  $d^*(T)$  is a statistic (does not depend on  $\theta$ ).
  - (b)  $d^*(T)$  is an unbiased estimator of  $\tau(\theta)$ .
  - (c)  $\text{Var}(d^*(T)) \leq \text{Var}(d(\mathbf{X}))$  for all  $\theta$ . Equality holds iff  $d(\mathbf{X}) = d^*(T)$  a.s.
- **Theorem 5.4:** Any MLE is a function of a sufficient statistic.

# Lecture 6: Minimal Sufficiency and Completeness

## 6.1 Minimal Sufficiency

- **Definition 6.1:** A sufficient statistic  $T$  is minimal sufficient if it is a function of any other sufficient statistic  $S$ .
- **Theorem 6.1 (Lehmann-Scheffé):**  $T(\mathbf{x})$  is minimal sufficient if for any two sample points  $\mathbf{x}$  and  $\mathbf{y}$ , the ratio  $L(\theta|\mathbf{x})/L(\theta|\mathbf{y})$  is independent of  $\theta$  if and only if  $T(\mathbf{x}) = T(\mathbf{y})$ .

## 6.2 Completeness

- **Definition 6.2:** A statistic  $T$  is complete if  $E_\theta[h(T)] = 0$  for all  $\theta$  implies  $P_\theta(h(T) = 0) = 1$  for all  $\theta$ . (Completeness means the only unbiased estimator of zero is zero itself).
- **Theorem 6.2:** If  $X_1, \dots, X_n$  are i.i.d. from a **full-rank Exponential Family** with the natural form:

$$f(x|\eta) = h(x) \exp \{ \eta T(x) - A(\eta) \}$$

The natural statistic  $\mathbf{T}(\mathbf{X})$  is **complete and sufficient**. And, the joint density of the sample  $\mathbf{X}$  is also an Exponential Family:

$$f(\mathbf{x}|\eta) = \left[ \prod_{i=1}^n h(x_i) \right] \exp \left\{ \eta \sum_{i=1}^n T(x_i) - nA(\eta) \right\}$$

The natural sufficient statistic  $\mathbf{T}(\mathbf{X}) = \sum_{i=1}^n \mathbf{T}(\mathbf{x}_i)$  is **complete and sufficient**.

## 6.3 Lehmann-Scheffé Theorem (Finding UMVUE)

- **Theorem 6.3:** If  $T$  is a **complete sufficient** statistic and  $g(T)$  is an **unbiased** estimator of  $\tau(\theta)$ , then  $g(T)$  is the **unique UMVUE**.
- **Remark:** This provides the “Gold Standard” for finding UMVUE:
  1. Find a complete sufficient statistic  $T$  (usually via EF).
  2. Find a function  $g(T)$  such that  $E[g(T)] = \tau(\theta)$ .

# Lecture 7: Hypothesis Testing: Basic Concepts

## 7.1 Statistical Hypotheses

- **Null Hypothesis ( $H_0$ ):** The statement being tested (usually status quo).
- **Alternative Hypothesis ( $H_1$ ):** The statement we hope to find evidence for.
- **Simple Hypothesis:** Specifies the distribution completely (e.g.,  $H_0 : \theta = \theta_0$ ).
- **Composite Hypothesis:** Does not specify the distribution completely (e.g.,  $H_1 : \theta > \theta_0$ ).

## 7.2 Types of Errors

- **Type I Error:** Rejecting  $H_0$  when it is true.  $P(\text{Type I Error}) = \alpha$ .
- **Type II Error:** Failing to reject  $H_0$  when it is false.  $P(\text{Type II Error}) = \beta$ .
- **Power Function  $K(\theta)$ :**  $P_\theta(\text{Reject } H_0)$ .
  - For  $\theta \in \Theta_0$ ,  $K(\theta) = P(\text{Type I Error})$ .
  - For  $\theta \in \Theta_1$ ,  $K(\theta) = \text{Power of the test} = 1 - \beta(\theta)$ .
- **Size ( $\alpha$ ):** The supremum of the Type I error probability:  $\alpha = \sup_{\theta \in \Theta_0} K(\theta)$ .
- **Level:** A test is of level  $\alpha$  if its size is  $\leq \alpha$ .

## 7.3 Neyman-Pearson Lemma (Simple vs. Simple)

- **Theorem 7.1:** For testing  $H_0 : \theta = \theta_0$  vs  $H_1 : \theta = \theta_1$ , the most powerful (MP) test of size  $\alpha$  is the Likelihood Ratio Test:
  - Reject  $H_0$  if  $\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} > k$ .
  - Fail to reject if  $\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} < k$ .
  - (Randomize if necessary for discrete cases).

## 7.4 P-Value

- **Definition:** The smallest level  $\alpha$  at which the observed data would lead to rejection of  $H_0$ .
- **Property:** If the p-value  $\leq \alpha$ , reject  $H_0$  at level  $\alpha$ .

# Lecture 8: Composite Hypothesis Testing & MLR

## 8.1 Randomized Tests

- For discrete distributions, an exact size  $\alpha$  might not be achievable with a simple threshold.
- **Randomized Test Function:**  $\phi(\mathbf{X}) = P(\text{Reject } H_0|\mathbf{X})$ .
  - e.g., Reject if  $T > c$ , reject with probability  $\gamma$  if  $T = c$ , fail to reject if  $T < c$ .

## 8.2 Monotone Likelihood Ratio (MLR)

- **Definition 8.1:** A family of PDFs  $\{f(x|\theta)\}$  has MLR in a statistic  $T(\mathbf{X})$  if for any  $\theta_2 > \theta_1$ , the ratio  $L(\theta_2|\mathbf{X})/L(\theta_1|\mathbf{X})$  is a non-decreasing function of  $T(\mathbf{X})$ .
- **Exponential Family:** If  $f(x|\theta) = h(x) \exp[Q(\theta)T(x) - A(\theta)]$  and  $Q(\theta)$  is strictly increasing, then the family has MLR in  $T(\mathbf{X})$ .

## 8.3 Uniformly Most Powerful (UMP) Tests

- **Theorem 8.1:** If a family has MLR in  $T(\mathbf{X})$ , then for testing  $H_0 : \theta \leq \theta_0$  vs  $H_1 : \theta > \theta_0$ , there exists a UMP test of size  $\alpha$  given by:

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } T(\mathbf{x}) > c \\ \gamma & \text{if } T(\mathbf{x}) = c \\ 0 & \text{if } T(\mathbf{x}) < c \end{cases}$$

- **Remark:** UMP tests usually exist for one-sided alternatives in MLR families but rarely exist for two-sided alternatives ( $H_1 : \theta \neq \theta_0$ ).

# Lecture 9: Likelihood Ratio Test and Others

## 9.1 Generalized Likelihood Ratio Test (LRT)

- **Setup:** Test  $H_0 : \theta \in \Theta_0$  vs  $H_1 : \theta \in \Theta_1$ , where at least one hypothesis is composite. Unless  $\Theta_0$  has special structure, no UMP/UMPU test typically exists.
- **Generalized LRT Statistic (9.1):**

$$\lambda(\mathbf{X}) = \frac{\sup_{\theta \in \Theta_0} L(\theta|\mathbf{X})}{\sup_{\theta \in \Theta_0 \cup \Theta_1} L(\theta|\mathbf{X})}$$

Reject  $H_0$  if  $\lambda \leq \lambda_0$ , where  $\lambda_0$  yields the required test size.

- **Remarks:**

1.  $0 \leq \lambda \leq 1$  (numerator  $\geq 0$ ; denominator's parameter space contains the numerator's).

2. The generalized LRT is **NOT** equivalent to the Neyman-Pearson MP test even when both hypotheses are simple — so  $\lambda$  does not necessarily yield the most powerful test.
3. The denominator typically uses the unconstrained MLE  $\hat{\theta}$ ; the numerator uses the constrained MLE  $\check{\theta}$  under  $H_0$ .

## 9.2 Key LRT Examples

- **Example 9.1 (Two-sided  $t$ -test):** For  $X_i \sim N(\mu, \sigma^2)$ , testing  $H_0 : \mu = \mu_0$ , the LRT yields:

$$\lambda(\mathbf{X}) = \left( \frac{1}{1 + t^2/(n-1)} \right)^{n/2}, \quad t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Since  $\lambda$  is monotone decreasing in  $t^2$ , rejecting  $\lambda < \lambda_0$  is equivalent to rejecting when  $|t| > t_{n-1, \alpha/2}$ .

- **Example 9.2 (Large-sample one-sided test for exponential rate):** For  $f(x|\theta) = \theta e^{-\theta x}$ ,  $x > 0$ , testing  $H_0 : \theta \leq \theta_0$ , the LRT reduces to rejecting when  $y = \theta_0 \bar{X} < k$  for some  $k < 1$ . Using CLT,  $\sqrt{n}(\theta \bar{X} - 1) \xrightarrow{d} N(0, 1)$ , so the large-sample test rejects  $H_0$  if:

$$\sqrt{n}(\theta_0 \bar{X} - 1) < -z_\alpha$$

## 9.3 Proper Hypothesis

- **Definition 9.1:** A null hypothesis is **proper** if  $\Theta_0$  can be characterized either as:
  - **(9.2) Restriction form:**  $H_0 : \mathbf{R}(\theta) = \mathbf{0}$  vs  $H_1 : \mathbf{R}(\theta) \neq \mathbf{0}$ , where  $\mathbf{R} : \mathbb{R}^d \rightarrow \mathbb{R}^r$  is differentiable and the Jacobian  $\mathbf{C}(\theta)_{r \times d} = \partial \mathbf{R} / \partial \theta$  has rank  $r$ , with  $0 < r \leq d$ .
  - **(9.3) Reparameterization form:**  $H_0 : \theta = \mathbf{g}(\psi)$  vs  $H_1 : \theta \neq \mathbf{g}(\psi)$ , where  $\mathbf{g} : \Psi \rightarrow \mathbb{R}^d$  is injective, differentiable,  $\Psi \subset \mathbb{R}^{d-r}$  open, and  $\mathbf{D}(\psi)_{d \times (d-r)} = \partial \mathbf{g} / \partial \psi$  has rank  $d - r$ .

## 9.4 Three Asymptotic Test Statistics

Let  $\hat{\theta}_n$  = unconstrained MLE;  $\check{\theta}_n$  = constrained MLE under  $H_0$ ;  $i_n(\theta)$  = Fisher Information;  $S(\theta|\mathbf{X})$  = score.

- **Definition 9.2:** Three test statistics:
  1. **Likelihood Ratio (LRT):**  $Q_L = 2[\log L(\hat{\theta}_n|\mathbf{X}) - \log L(\check{\theta}_n|\mathbf{X})] = -2 \log \lambda(\mathbf{X})$ . (uses both MLEs)
  2. **Wald's Statistic:**  $Q_W = \mathbf{R}^T(\hat{\theta}_n) \left\{ \mathbf{C}(\hat{\theta}_n) i_n^{-1}(\hat{\theta}_n) \mathbf{C}^T(\hat{\theta}_n) \right\}^{-1} \mathbf{R}(\hat{\theta}_n)$ . (uses only unconstrained MLE)
  3. **Rao's Score Statistic:**  $Q_R = S^T(\check{\theta}_n|\mathbf{X}) i_n^{-1}(\check{\theta}_n) S(\check{\theta}_n|\mathbf{X})$ . (uses only constrained MLE)
- **Theorem 9.1 (Asymptotic Equivalence):** Under regularity conditions and  $H_0$  as in (9.2) or (9.3):

$$Q_L, Q_W, Q_R \xrightarrow{d} \chi_r^2$$

Hence each test rejects  $H_0$  when  $Q > \chi_{r, \alpha}^2$  for an asymptotic size- $\alpha$  test. Here  $r$  is the number of restrictions in  $\mathbf{R}(\theta) = \mathbf{0}$ .

- **Example 9.3 (Joint test on normal):** For  $X_i \sim N(\mu, \sigma^2)$ , test  $H_0 : \mu = 0, \sigma^2 = 1$ . With  $\mathbf{R}(\theta) = (\mu, \sigma^2 - 1)^T$ ,  $\partial \mathbf{R} / \partial \theta = I_2$  has rank  $r = 2$ , so  $-2 \log \lambda(\mathbf{X}) \xrightarrow{d} \chi_2^2$ .

# Lecture 10: Bayesian Statistics and Statistical Decision

## 10.1 Bayesian Framework

- **Bayesian Paradigm:**  $\theta$  is no longer a fixed constant but a random variable with a **prior distribution**  $\pi(\theta)$  chosen before data collection.
- **Bayes' Theorem (10.1) (Posterior):**

$$\pi(\theta|\mathbf{X}) = \frac{f(\mathbf{X}|\theta)\pi(\theta)}{\int f(\mathbf{X}|\theta)\pi(\theta)d\theta}$$

In short: **posterior**  $\propto$  **prior**  $\times$  **likelihood** (10.2). The denominator is the marginal density  $f(\mathbf{X}) = \int f(\mathbf{X}|\theta)\pi(\theta)d\theta$ .



- **Conjugate Prior:** If  $\pi(\theta)$  and  $\pi(\theta|\mathbf{X})$  are from the same family,  $\pi(\theta)$  is a conjugate prior w.r.t.  $f(\mathbf{X}|\theta)$ . Common pairs:
  - Beta prior + Binomial likelihood  $\Rightarrow$  Beta posterior.
  - Normal prior + Normal likelihood (known  $\sigma^2$ )  $\Rightarrow$  Normal posterior.
  - Gamma prior + Poisson likelihood  $\Rightarrow$  Gamma posterior.

## 10.2 Bayesian Hypothesis Testing

- **Posterior probability test:** For  $H_0 : \theta \geq \theta_0$  vs  $H_1 : \theta < \theta_0$ , base test on  $P(\theta \geq \theta_0|\mathbf{X})$ . Larger value = more support for  $H_0$ .
- **Bayes Factor (BF):** Useful for point hypotheses where  $P(\theta = \theta_0|\mathbf{X}) = 0$  under continuous priors. With prior probabilities  $\pi_0 = P(\theta \in \Theta_0)$ ,  $\pi_1 = 1 - \pi_0$ , and posterior probabilities  $p_0, p_1$ :

$$\text{BF} = \frac{\text{posterior odds of } H_0}{\text{prior odds of } H_0} = \frac{p_0/p_1}{\pi_0/\pi_1}$$

- **Simple vs. Simple (10.3):** For  $H_0 : \theta = \theta_0$  vs  $H_1 : \theta = \theta_1$ , the Bayes Factor equals the **likelihood ratio**:

$$\text{BF} = \frac{f(\mathbf{X}|\theta_0)}{f(\mathbf{X}|\theta_1)}$$

- **Composite Hypotheses:** With normalized priors  $\rho_i(\theta) = \pi(\theta)/\pi_i$  on  $\Theta_i$ :

$$\text{BF} = \frac{\int_{\Theta_0} f(\mathbf{X}|\theta)\rho_0(\theta)d\theta}{\int_{\Theta_1} f(\mathbf{X}|\theta)\rho_1(\theta)d\theta}$$

- **Jeffreys' Scale (Support for  $H_0$ ):** | BF | Support for  $H_0$  | |—|—| |  $< 1$  | Negative | | 1–3 | Slight | | 3–12 | Positive | | 12–150 | Strong | |  $> 150$  | Very strong |

## 10.3 Posterior Bayes Estimator

- **Definition:** The **posterior Bayes estimator** of  $\tau(\theta)$  is the posterior mean:

$$E[\tau(\theta)|\mathbf{X}] = \int \tau(\theta)\pi(\theta|\mathbf{X})d\theta$$

- **Example 10.1 (Normal-Normal):**  $X_i \sim N(\mu, \sigma_0^2)$  with prior  $\mu \sim N(\xi, \tau^2)$ :

$$\pi(\mu|\mathbf{X}) \sim N(A, B^2), \quad A = \frac{\frac{n\bar{X}}{\sigma_0^2} + \frac{\xi}{\tau^2}}{\frac{n}{\sigma_0^2} + \frac{1}{\tau^2}}, \quad B^2 = \frac{1}{\frac{n}{\sigma_0^2} + \frac{1}{\tau^2}}$$

The posterior mean  $A$  is a precision-weighted average of the prior mean and the data mean. For large  $n$ , prior choice matters little.

- **Example 10.4 (Bernoulli with Uniform prior):**  $X_i$  i.i.d. Bernoulli( $\theta$ ), prior  $\theta \sim \text{Unif}(0, 1) = \text{Beta}(1, 1)$ . Posterior is Beta( $\sum X_i + 1, n - \sum X_i + 1$ ), so:

$$E[\theta|\mathbf{X}] = \frac{\sum X_i + 1}{n + 2}$$

This Bayes estimator is biased (the MLE  $\bar{X}$  is unbiased/UMVUE — Bayes estimators are generally biased).

## 10.4 Statistical Decision Theory: Loss and Risk

- **Decision Function:** An estimator  $T(\mathbf{X})$  viewed as a decision function;  $T(\mathbf{x})$  is the decision/estimate upon observing  $\mathbf{X} = \mathbf{x}$ .
- **Loss Function  $L(\theta, T)$ :** Must satisfy  $L(\theta, T) \geq 0$  and  $L(\theta, T) = 0 \iff T = \tau(\theta)$ .
  - **Quadratic loss:**  $L(\theta, T) = (T - \tau(\theta))^2$ .
  - **Absolute loss:**  $L(\theta, T) = |T - \tau(\theta)|$ .
  - $\epsilon$ -loss:  $L(\theta, T) = A \cdot \mathbf{1}\{|T - \tau(\theta)| > \epsilon\}$ .
- **Risk Function (10.4):**  $R_T(\theta) = E_\theta[L(\theta, T(\mathbf{X}))]$ . Under quadratic loss,  $R_T(\theta) = \text{MSE}(T)$ .

## 10.5 Admissibility and Dominance

- **Inadmissible:**  $T^*$  is **inadmissible** if there exists  $T$  such that:

1.  $R_T(\theta) \leq R_{T^*}(\theta)$  for all  $\theta \in \Theta$ , AND
2. strict inequality holds for at least one  $\theta$ .

Then  $T$  **dominates**  $T^*$  (or  $T$  is **better than**  $T^*$ ). If only (1) holds,  $T$  is **as good as**  $T^*$ .

- **Admissible:**  $T^*$  is admissible if it is not inadmissible. Admissibility is **necessary but not sufficient** for being optimal/best (best = as good as every other estimator).

## 10.6 Bayes Risk and Bayes Estimators

- **Bayes Risk (10.5):** Average risk over the prior:

$$r(T) = \int R_T(\theta)\pi(\theta)d\theta$$

- **Bayes Estimator:**  $T^* = \arg \min_T r(T)$ . **A unique Bayes estimator is always admissible.**
- **Theorem 10.1 (Key):** The Bayes estimator minimizes the **posterior expected loss**:

$$T^*(\mathbf{x}) = \arg \min_a \int L(\theta, a)\pi(\theta|\mathbf{x})d\theta$$

*Proof sketch:* swap order of integration in  $r(T) = \int \int L(\theta, T(\mathbf{x}))f(\mathbf{x}|\theta)\pi(\theta)d\mathbf{x}d\theta$ , then minimize the inner integral pointwise in  $\mathbf{x}$ .

- **Corollary 10.1 (Bayes estimators by loss):**

1. **Quadratic loss:**  $T^*(\mathbf{X}) = E[\tau(\theta)|\mathbf{X}]$  (posterior **mean**).
2. **Absolute loss:**  $T^*(\mathbf{X}) =$  posterior **median** of  $\tau(\theta)|\mathbf{X}$ .

- **Example 10.7:**  $X \sim \text{Unif}(0, \theta)$ , prior  $\pi(\theta) = \theta e^{-\theta}$ . The posterior is  $\theta|X \sim X + \text{Exponential}(1)$ , so under absolute loss the Bayes estimator is  $T^* = X + \log 2$  (since the median of  $\text{Exp}(1)$  is  $\log 2$ ).

## 10.7 Minimax Estimator

- **Definition:**  $T^*$  is **minimax** if  $\max_{\theta} R_{T^*}(\theta) \leq \max_{\theta} R_T(\theta)$  for all  $T$ . It minimizes the worst-case (maximum) risk.
- **Theorem 10.2:** If a Bayes estimator has **constant risk** (independent of  $\theta$ ), then it is also minimax.
- **Example 10.9 (Minimax for Binomial):**  $X \sim \text{Bin}(n, \theta)$ , prior  $\theta \sim \text{Beta}(\alpha, \beta)$ . Posterior is  $\text{Beta}(\alpha + x, \beta + n - x)$ , and the Bayes estimator (quadratic loss) is  $T = (X + \alpha)/(n + \alpha + \beta)$ . Choosing  $\alpha = \beta = \sqrt{n}/2$  makes the risk constant in  $\theta$ , so:

$$T^* = \frac{X + \sqrt{n}/2}{n + \sqrt{n}}$$

is the minimax estimator of  $\theta$ .

## Calculation Reference Examples

- **Poisson( $\lambda$ ):**  $T = \sum X_i$  is complete sufficient. To estimate  $\tau(\lambda) = e^{-\lambda} = P(X_1 = 0)$ : We use  $d(\mathbf{X}) = I(X_1 = 0)$ . Then  $UMVUE = E[I(X_1 = 0) | \sum X_i = t] = P(X_1 = 0 | \sum X_i = t) = (1 - 1/n)^t$ .
- **Normal ( $\mu, \sigma^2$ ):**  $(\bar{X}, S^2)$  are jointly complete and sufficient.  $\bar{X} \perp S^2$  allows simple variance calculations for functions like  $\bar{X}^2 S^2$ .
- **Uniform(0,  $\theta$ ):**  $X_{(n)}$  is complete and sufficient (even though not EF).  $E[X_{(n)}] = \frac{n}{n+1}\theta \implies \frac{n+1}{n}X_{(n)}$  is the UMVUE of  $\theta$ .
- **Non-completeness Example:** For  $\text{Uniform}(\theta, \theta + 1)$ ,  $T = (X_{(1)}, X_{(n)})$  is minimal sufficient but NOT complete because  $E[X_{(n)} - X_{(1)}] = \frac{n-1}{n+1}$ , which is independent of  $\theta$ , so  $X_{(n)} - X_{(1)} - \frac{n-1}{n+1}$  is a non-zero function with zero expectation.

# Appendix: Mathematical Tools & Probability Identities

## A.1 Probability Laws

- **Conditional Probability:**  $P(A|B) = \frac{P(A \cap B)}{P(B)}$ .
- **Law of Total Probability:**  $P(A) = \sum_i P(A|B_i)P(B_i)$ .
- **Bayes' Rule:**  $P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$ .
- **Law of Total Expectation:**  $E[X] = E[E[X|Y]]$ .
- **Law of Total Variance:**  $Var(X) = E[Var(X|Y)] + Var(E[X|Y])$ .

## A.2 Calculus & Series

- **Taylor Expansion (around  $a$ ):**  $g(x) = g(a) + g'(a)(x-a) + \frac{g''(a)}{2!}(x-a)^2 + \dots$ 
  - Used in Delta Method and MLE asymptotic proofs.
- **Leibniz Integral Rule:**  $\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = f(b(t), t)b'(t) - f(a(t), t)a'(t) + \int_{a(t)}^{b(t)} \frac{\partial}{\partial t} f(x, t) dx$ .
  - Regularity conditions often assume  $\frac{d}{d\theta} \int f = \int \frac{\partial}{\partial \theta} f$ .

## A.3 Common Integrals

- **Gamma Function:**  $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$ . Note  $\Gamma(n) = (n-1)!$  and  $\Gamma(1/2) = \sqrt{\pi}$ .
- **Gamma Integral:**  $\int_0^\infty x^{\alpha-1} e^{-x/\beta} dx = \Gamma(\alpha)\beta^\alpha$ .
- **Beta Function:**  $B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ .
- **Normal Integral:**  $\int_{-\infty}^\infty e^{-ax^2+bx+c} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}+c}$ .
- **Integration by Parts:**  $\int u dv = uv - \int v du$ . Useful for finding moments.